

Let's Create Layer by Layer: Image Generation via Next-Layer Prediction

02501 Advanced Deep Learning in Computer Vision
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1 Project Description

Autoregressive (AR) GPT-like models have achieved remarkable success in text generation, inspiring researchers to extend similar AR principles to image generation. However, tokenizing images is not straightforward, leading to different design choices as can be seen in Fig. 1. Traditional AR image generation models produce images sequentially in a patch-wise manner, with some approaches following a raster-scan order (left to right, top to bottom). In contrast, Visual Autoregressive (VAR) models [1] generate multi-scale token maps autoregressively, progressing from coarse to fine scales (lower to higher resolutions).

In this project, we propose a novel **layer-wise autoregressive model**, which constructs images hierarchically from coarse background elements to fine detailed objects. Instead of predicting patch sequences or entire images at once, our approach mimics the human-like artistic process—starting with the background and progressively adding objects and finer details. This formulation provides a more structured and interpretable decomposition of the image generation process.

Four Different Autoregressive Models

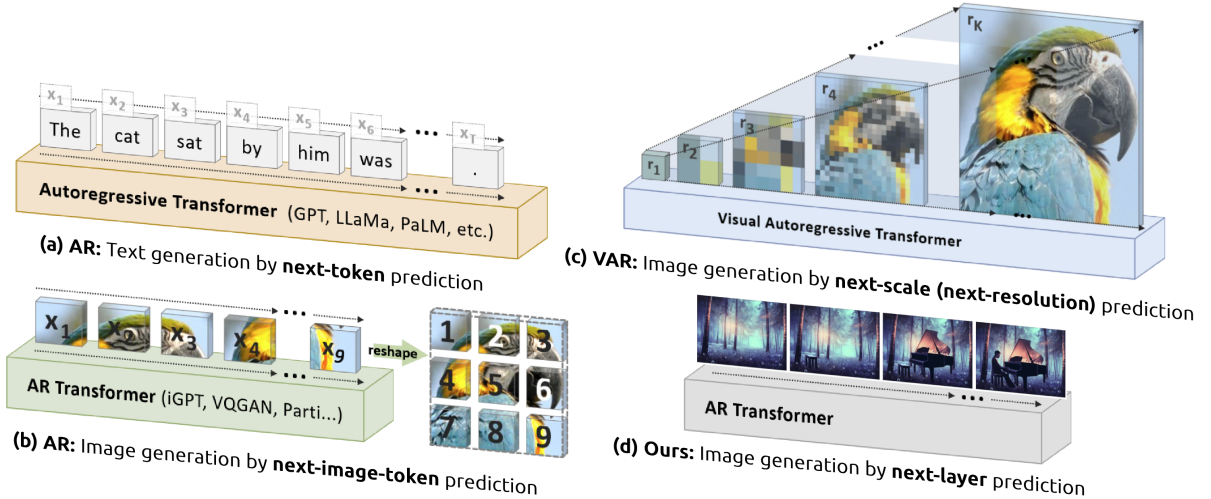


Figure 1: (a) AR applied to language: sequential text token generation from left to right, word by word; (b) AR applied to images: sequential visual token generation in a raster-scan order, from left to right, top to bottom; (c) VAR for images: multi-scale token maps are autoregressively generated from coarse to fine scales (lower to higher resolutions), with parallel token generation within each scale. (d) Ours: multi-layered images are autoregressively generated from background to fine details.

Our method differs fundamentally from diffusion models. While diffusion models break the image generation problem into small denoising steps [2] and VAR splits it into next-scale predictions, our approach splits it into next layer prediction subproblems—progressing from background to foreground objects in a natural hierarchy.

This approach offers a more natural decomposition of the image generation process, making it valuable for various applications. We will evaluate its efficiency and effectiveness compared to existing methods, including diffusion models and VAR.

2 Tasks

In this project, you could work on the following tasks:

Task 1: Dataset Creation. Construct a synthetic dataset tailored for layer-wise autoregressive image generation. First, collect individual PNG assets to allow flexible dataset construction. Then, generate a large number of possible combinations to create a diverse dataset. The dataset should consist of low-resolution images to enable efficient training. You can choose one of the following structured setups: (i) Character generation, where images are built from modular components such as a base face, different eyes, noses, and other features¹; (ii) Pizza generation, where images consist of different bases, toppings, and sauces; (iii) Chessboard configurations, where variations include different background colors and diverse piece placements and styles.

Task 2: Model Implementation. Implement the proposed layer-wise autoregressive model and train it on the generated small synthetic low-resolution dataset. Ensure the model effectively learns to generate images progressively from background to foreground.

Task 3: Evaluation. Assess the generation quality and generalization capabilities. Evaluate how well the model reconstructs images layer by layer and analyze its ability to generate plausible and diverse compositions. Explain your design choices.

Task 4: Comparison with Diffusion Models. Train a diffusion model using the same number of parameters and inference budget as the proposed model. Compare the performance in terms of image quality and generation speed.

Task 5: Comparison with VAR. Train a VAR model with an equivalent setup, where images are generated through next-scale predictions. Evaluate its performance relative to the proposed approach in terms of structured generation and overall quality.

Task 6: Large-Scale Training on the MULAN Dataset. Train the model on the MULAN dataset [3], which consists of multi-layered images depicting structured compositions (as in Fig. 2). Repeat the comparisons with diffusion and VAR models in this large-scale setting. Since high-resolution image generation is computationally expensive, consider working in the latent space using a pretrained variational autoencoder (VAE) to improve efficiency.



Figure 2: Samples from the MULAN dataset that can be used for training our model. Each row illustrates a progressive image composition process, where new elements are added to a scene step by step.

¹You can use these free assets: <https://kenney.nl/assets/modular-characters> different

References

- [1] K. Tian, Y. Jiang, Z. Yuan, B. PENG, and L. Wang, “Visual autoregressive modeling: Scalable image generation via next-scale prediction,” in *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- [2] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” *Advances in neural information processing systems*, vol. 33, pp. 6840–6851, 2020.
- [3] P.-D. Tudosiu, Y. Yang, S. Zhang, *et al.*, “Mulan: A multi layer annotated dataset for controllable text-to-image generation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2024, pp. 22 413–22 422.